Gi Joo Kim, Ph.D.

Research Scientist · ByWater Institute at Tulane University
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Education

Seoul National University, Seoul, South Korea
- Ph.D. in Civil and Environmental Engineering — *Mar 2017 – Feb 2022*
  (Dissertation Title: Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models)

Korea University, Seoul, South Korea
- B.S. in Architectural, Civil, and Environmental Engineering — *Mar 2013 – Feb 2017*

Academic Appointments

Research Scientist
- ByWater Institute, Tulane University, New Orleans, LA, USA — *Sep 2025 – Present*

Postdoctoral Scholar
- Lamontagne Lab, Tufts University, Medford, MA, USA — *Jun 2023 – Sep 2025*
  Advisor: Dr. Jonathan Lamontagne
- Institute of Construction and Environmental Engineering, Seoul National University, Seoul, South Korea — *Mar 2022 – Feb 2023*
  Advisor: Dr. Young-Oh Kim

Lecturer
- Department of Civil and Environmental Engineering, Seoul National University, Seoul, South Korea — *Sep 2022 – Feb 2023*

Visiting Student
- Reed Research Group, Cornell University, Ithaca, NY, USA — *Sep 2019 – Dec 2019*
  Advisor: Dr. Patrick Reed

Academic Publications

A. Peer-Reviewed Journal Articles *(n=7; * indicates corresponding author)*

Kim, G.J.*, O'Neill, B., Morris, J., Wise, M., Weyant, J., & Lamontagne, J.R. (2026). Identifying key uncertainties and drivers of future resource security outcomes through a multisector scenario ensemble. *Earth's Future*, 14, e2025EF007585. https://doi.org/10.1029/2025EF00758

5 (Media Coverage: *MIT News*)

Chowdhury, A.F.M.K.*, Abeshu, G.W., Zhao, M., Wild, T.B., Niazi, H., Zhang, Y., **Kim, G.J.,** Binsted, M., Lamontagne, J., Li, H.-Y. (in press). Hydropower Site-level Monthly Availability Data Ensemble for 1950-2100 at Existing and Potential Global Sites. *Scientific Data.*

Kim, G.J.*, Wessel, J., Birnbaum, A., Moraites, G., Snyder, A., Morris, J., Wild, T.B., Lamontagne, J.R. (2025). Discovering the multisectoral impacts of global energy sector outcomes through multiple ensemble aggregation measures. *Earth's Future*, 13, e2025EF006526. https://doi.org/10.1029/2025EF006526

Kim, G.J., Kim, D.H., & Kim, Y.-O.* (2024). Improving probabilistic drought prediction with soil moisture information under the ensemble streamflow prediction framework. *Stochastic Environmental Research and Risk Assessment*, 38, 2767-2779. https://doi.org/10.1007/s00477-024-02710-6

Kim, G.J., Seo, S.B., & Kim, Y.-O.* (2022). Adaptive Reservoir Management by Reforming the Zone-based Hedging Rules against Multi-year Droughts. *Water Resources Management*, 36, 3575-3590. https://doi.org/10.1007/s11269-022-03214-0

Kim, G.J., Kim, Y.-O.*, & Reed, P.M. (2021). Improving the robustness of reservoir operations with stochastic dynamic programming. *Journal of Water Resources Planning and Management,* 147(7), 04021030. https://doi.org/10.1061/(ASCE)WR.1943-5452.0001381

Kim, G.J., & Kim, Y.-O.* (2021). How does the coupling of real-world policies with optimization models expand the practicality of solutions in reservoir operation problems? *Water Resources Management*, 35, 3121-3137. https://doi.org/10.1007/s11269-021-02862-y

B. Papers In Revision & Under Review

Kim, G.J.*, Wild, T.B., Zhang, Y., Binsted, M., Niazi, H., Chowdhury, K., Zhao, M., & Lamontagne, J.R. (in review). Beyond Water: Global Hydropower Futures Shaped by Interacting Hydrological and Socio-Technical Drivers. *Nature Water.* (Link to preprint: https://doi.org/10.21203/rs.3.rs-9464688/v1)

Kim, G.J.*, Karamyar, F., Sefair, J.A. & Sabo, J. L. (submitted). Robust River Flow Design Balances Food-Energy-Water Tradeoffs in the Lower Mekong Basin. *Nature Communications.*

C. Papers in Preparation

Kim, G.J.*, O'Neill, B., Morris, J., Weyant, J., Lamontagne, J.R., & Sabo, J.L. (in prep). Designing Regionally Informed Global Marker Scenarios for Subregional Human Well-Being. *Earth's Future.* (Estimated Submission: Nov 2026)

Koval, D.K.*, **Kim, G.J.**, Thompson, I., & Lamontagne, J.R. (in prep). Designing Global Hydrologic Marker Scenarios for Water Scarcity and Food Burden. *Earth's Future.* (Estimated Submission: Mar 2027)

Choi, S.H., **Kim, G.J.**, Kim, Y.-O. (in prep). Integrating Deep Learning for Streamflow Prediction with Reinforcement Learning for Coordinated Reservoir Operation. *Water Resources Research*. (Estimated Submission: Mar 2027)

Chowdhury, A.F.M.K., Niazi, H., Zhang, Y., Binsted, M., Abeshu, G.W., Zhao, M., **Kim, G.J.,** Lamontagne, J., and Li, H.-Y. (in prep). GCAM-hydropower v2.0: enhanced representation of global hydropower in a global multi-sector model. *Geoscientific Model Development*.

D. Peer-Reviewed Journal Articles - KCI *(Abstracts in English, Text in Korean)*

Lee, J.H., **Kim, G.J.**, & Kim, Y.-O.* (2024). A Study on the Feasibility Analysis of the Current Flood Season: A Case Study of the Yongdam Dam. *Journal of Korea Water Resources Association*, 57(5), 359-369. https://doi.org/10.3741/JKWRA.2024.57.5.359

Kim, Y., **Kim, G.J.**, & Kim, Y.-O.* (2023). Analysis of Ensemble Streamflow Prediction Effect on Deriving Dam Releases for Water Supply. *Journal of Korea Water Resources Association,* 56(12), 969-980. https://doi.org/10.3741/JKWRA.2023.56.12.969

Kim, G.J., Kim, J.*, Seo, S.B., & Kim, Y.-O. (2023). Development of an Evaluation Index based on Supply Capacity for Realistic Drought Resilience Evaluation. *Journal of Korea Water Resources Association,* 56(1), 11-21. https://doi.org/10.3741/JKWRA.2023.56.1.11

Kim, G.J.*, Seo, S.B., & Kim, Y.-O. (2019). Elicitation of drought alternatives based on Water Policy Council and the role of Shared Vision Model. *Journal of Korea Water Resources Association*, 52(6), 429-440. https://doi.org/10.3741/JKWRA.2019.52.6.429

E. Books and Reports

Kim, Y.-O., Seo, S.B., & **Kim, G.J.** (2020). Incorporation of robustness and adaptiveness into reservoir operations under climate change. In Teegavarapu, R.S.V., Kolokytha, E., & Oliveria Galvão, C. (Ed.) Climate Change-Sensitive Water Resources Management (pp.140-160). London, UK: CRC Press.

Research Funding & Grants

A. Secured Research Funding & Grants as PI(s)

Total funding secured as principal investigator to date: $41,486

P.I., 2024-2025	"**Development of Multisector Scenarios to Evaluate Societal well-being and Resilience"** PI G.J. Kim (Tufts University). National Research Foundation of Korea, Postdoctoral Fellowship Program	*$41,486*

B. Other Research Projects Involved

S.P., 2027-2030	**"Engineering Nature 2.0 to boost the performance of existing built infrastructure in protecting people from water-related natural disasters"** PI K.S. Candan (Arizona State University), US Army Corps of Engineers
Contributor, 2023-2025	**"Global Change Intersectoral Modeling System (GCIMS)"** PI M. Wise (Pacific Northwest National Laboratory), US Department of Energy, Office of Science, Earth and Environmental Systems Modeling program
Contributor, 2021-2022	**"Relief of water shortage in western Chungcheongnam-do by regenerating inflow and improving water supply adjustment standards"** PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
Contributor, 2020-2022	**"Study on the assessment of drought response capability and the improvement plan of coordinated dam-weir operation for river systems"** PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
Contributor, 2019-2020	**"Development of Drought Information Producing Technology for the Public"** PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
Contributor, 2017-2019	**"Climate Change Adaptation for Water Resources (CCAW)"** PI D.-H. Bae (Sejong University), Ministry of Environment, Republic of Korea

Awards & Honors

Jan 2023	**2023 Best Ph.D. Dissertation Award,** Korea Water Resources Association
Aug 2022	**Public Water & Big Data Competition Award**, Korea Water Resources Association
Feb 2022	**Young Member Award,** Gwanak Water Engineering Committee
Jun 2021	**Best Presentation Award,** 2021 Korea Water Resources Association Conference
Jul 2018	**Excellence Award**, 2018 Korea Climate Technology Idea Proposal Competition
May 2018	**Best Presentation Award**, 2018 Korea Water Resources Association Conference
Jul 2017	**Excellence Award**, Climate Change Academy for College Students

Invited & Conference Presentations

A. Invited Talks

Kim, G.J. (2026). Identifying Key Drivers of Future Food, Energy, and Water Security Outcomes through a Multisector Scenario Ensemble. Department of Civil & Environmental Engineering. Seoul National University, Seoul, South Korea. May 12, 2026.

Kim, G.J. (2026). Modeling Human-Earth System Interactions under Global Environmental Change. Department of Civil Engineering. Hannam University, Daejeon, South Korea. Apr

30, 2026.

Kim, G.J. (2025). From Global Resource Security to Local Water Resources Management: Modeling Human-Earth System Interactions under Climate Change. Department of River-Costal Science and Engineering. Tulane University, New Orleans LA, USA. Nov 6, 2025.

Kim, G.J. (2022). Adaptive and Robust Reservoir Management under Climate Change. Department of Civil, Environmental, and Plant Engineering, Chung-Ang University, Seoul, South Korea. Nov 16, 2022.

B. Conference Presentations

Kim, G.J., Karamyar, F., Sefair, J.A., Shah, R., & Sabo, J.L. (2026). Balancing Hydropower, Fisheries, and Rice Production Through Robust River Flow Design in the Lower Mekong Basin. AGU26.

Kim, G.J., Wessel, J., Lamontagne, J.R., Snyder, A., Wild, T.B., & Sabo, J.L. (2026). Aggregating Global Change Scenario Ensembles through Robustness Metrics. Decision Making Under Deep Uncertainty Society 2026 Annual Meeting.

Kim, G.J., Zhang, Y., Binsted, M., Niazi, H., Chowdhury, K., Zhao, M., Wild, T.B., & Lamontagne, J.R. (2025). Projecting Future Global and Regional Hydropower Expansion Considering Human and Earth System Interactions. AGU25.

Kim, G.J., O'Neill, B., Morris, J., Wise, M., Weyant, J., & Lamontagne, J.R. (2024). Identifying the Key Drivers of Global and Regional Food-Energy-Water Security Outcomes through Scenario Discovery. AGU24.

Wild, T.B., Graham, N.T., Zhao, M., Niazi, H., Chowdhury, A.F.M., Birnbaum, A.N., **Kim, G.J.**, et al. (2024). Representation of Reservoirs in Global Multi-Sector Models: Recent Advances and Future Directions. AGU24.

Kim, G.J., Lamontagne, J.R., Wild, T.B., Zhang, Y., Binsted, M., O'Neill, B., Morris, J., Wise, M., & Weyant, J. (2024). Large ensemble scenario discovery on global hydropower expansion and human well-being. 2024 EESM PI Meeting.

Kim, G.J., O'Neill, B., Lamontagne, J.R., Morris, J., Wise, M., & Weyant, J. (2024). Large ensemble scenario discovery on societal well-being and resilience. 2024 GCAM Annual Meeting.

Kim, G.J., Lee, J.H., Lee, J., & Kim, Y.-O. (2022). Adaptive Operation of Boryeong Dam Water Supply Adjustment Standards against Multi-year Droughts. 2022 Korea Water Resources Association Conference.

Kim, G.J., & Kim, Y.-O. (2021). Enhancing the Practical Applicability of Optimization Models with Real-world Policies in Reservoir Operation Problems. AOGS 2021.

Kim, G.J., & Kim, Y.-O. (2021). Comparative Analysis of Optimization Algorithms and the Effects of Coupling Hedging Rules in Reservoir Operations. 2021 Korea Water Resources Association Conference. (*Best Presentation Award*)

Kim, G.J., Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2019) Application of Shared Vision Planning for Drought Mitigation Climate Change Adaptation Council in Korea. 38th IAHR World Congress. https://doi.org/10.3850/38WC092019-1599

Kim, G.J., Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2019) Robust Dam Operations for Multi-year Drought Mitigation with Stakeholders in Korea. EWRI 2019.

Kim, G.J., Choi, S.H., Seo, S.B., & Kim, Y.-O. (2018). Development of a Shared Vision Planning Model for Participants' Consensus: Application in South Chungcheong Province, South Korea. The 36th International Conference of the System Dynamics Society, ISDC.

Kim, G.J., Choi, S.H., Seo, S.B., & Kim, Y.-O. (2018). Development of a Shared Vision Planning Model in Water Policy Council for Climate Change Adaptation in South Chungcheong Province. 2018 Korea Society of Climate Change International Conference.

Kim, G.J., Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2018). Boryeong Dam Inflow Time Series Generation that Reflects Multi-year Drought. 2018 Korea Water Resources Association Conference. (*Best Presentation Award*)

Yoon, H.N., **Kim, G.J.**, & Kim, Y.-O. (2018). Application of Robust Optimization for Reservoir Operation. 2017 Korea Water Resources Association Conference.

Teaching Experience

Instructor/Co-Instructor

- **Statistical Hydrology (457.568)** — *Sep 2022 – Feb 2023*
  Department of Civil and Environmental Engineering,
  Seoul National University, Seoul, South Korea
- **Water Resources Systems Engineering (457.557)** — *Mar 2022 – Aug 2022*
  Department of Civil and Environmental Engineering,
  Seoul National University, Seoul, South Korea

Graduate Student Mentoring

- Tufts University, Medford, MA, USA — *Sep 2023 – Aug 2025*
- Seoul National University, Seoul, South Korea — *Sep 2018 – Feb 2023*

Teaching Assistant

- **Hydrology, Water Resources Systems Engineering, Statistical Hydrology, Disaster and Risk Management, Water Resources Engineering** — *Sep 2018 – Feb 2022*
  Department of Civil and Environmental Engineering,
  Seoul National University, Seoul, South Korea

Professional Service and Memberships

A. Journal Peer Review

- *Nature Communications*
- *Earth's Future*
- *Water Resources Research*
- *Journal of Hydrology*
- *Natural Hazards*
- *Stochastic Environmental Research and Risk Assessment*
- *Journal of Water Resources Planning and Management*
- *Journal of Hydro-environmental Research*

B. Expert Panel Reviews and Advisory Activities

- Panel Reviewer. (2026). Gulf Research Program (Resilience Education), National Academies of Sciences, Engineering, and Medicine (NASEM).
- Expert Reviewer. (2026). IPCC Special Report on Climate Change and Cities (SRCITIES), Second Order Draft (SOD).

C. Professional Memberships

2026 - present **Decision Making Under Deep Uncertainty (DMDU) Society**, member
2024 – present **American Geophysical Union (AGU),** member
2019 – present **American Society of Civil Engineers (ASCE),** member
2019 – present **Asia Oceania Geosciences Society (AOGS),** member
2017 – present **Korea Water Resources Association (KWRA),** member